| **Exp. No: 9** | ANALYSIS OF PHOTODETECTOR NOISE AND SNR USING MATLAB |
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## OBJECTIVES :

1. Analyse the key noise sources in an optical photodetector, namely shot noise, dark current noise and thermal (receiver) noise.

2. Compute the individual and total noise current variances ($\sigma^2$) across a range of input optical powers.

3. Evaluate the Signal-to-Noise Ratio (SNR) and observe its scaling in the thermal noise-limited and shot-noise-limited regimes.

4. Calculate the Minimum Detectable Optical Power (Pmin) at SNR = 1 (0 dB).

5. Simulate and visualize photodetector noise and SNR curves using MATLAB.

## SOFTWARE REQUIRED :

• MATLAB

## MATLAB Program:

```
% Photodetector Noise and SNR Analysis

clc; clear;

close all;

%% 1. Constants and input parameters

q = 1.602e-19;  kB = 1.381e-23;  h = 6.626e-34;  c = 2.998e8;

lambda = 1550e-9;

eta    = 0.85;

Id     = 5e-9;

T      = 300;

RL     = 50;

Fn     = 2;

B      = 1e9;

Pdbm   = -60:0.1:0;

% wavelength (m)

% quantum efficiency

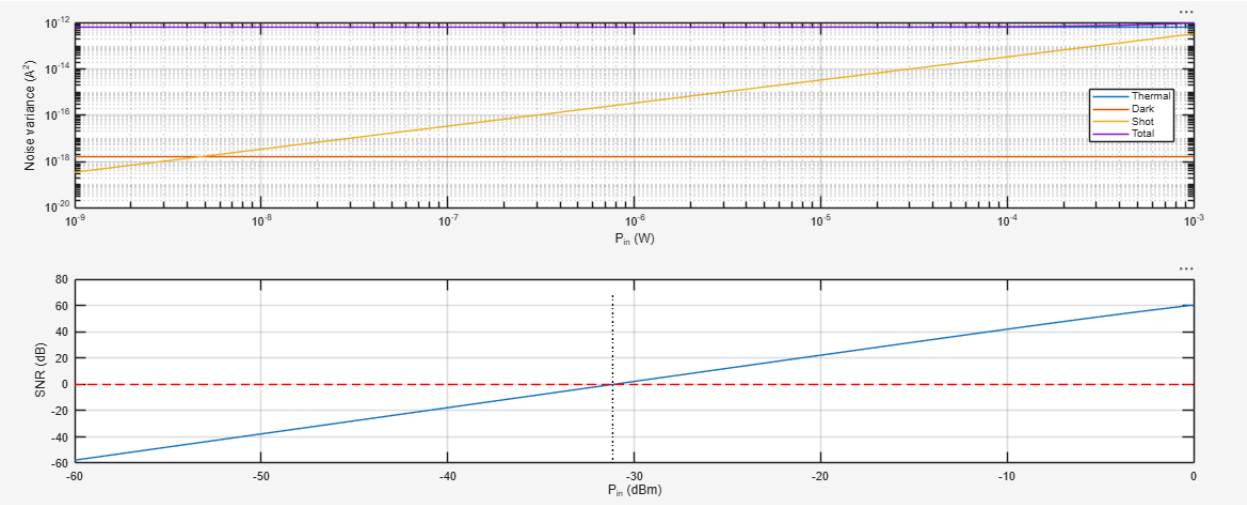
% dark current (A)

% temperature (K)
```

```
% load resistance (ohm)
% noise figure (3 dB = 2 linear)
% bandwidth (Hz)
% input power sweep (dBm)
P      = 10.^((Pdbm-30)/10);   % input power (W)
%% 2. Responsivity and signal current
R  = eta*q*lambda/(h*c);
Ip = R*P;
%% 3. Noise variances and SNR
var_shot = 2*q*Ip*B;
var_dark = 2*q*Id*B;
var_th   = 4*kB*T*B*Fn/RL;
var_tot  = var_shot + var_dark + var_th;
SNR_dB   = 10*log10(Ip.^2 ./ var_tot);
%% 4. Minimum detectable power (SNR = 1)
a = R^2;  b = -2*q*B*R;  cc = -(var_dark + var_th);
Pmin    = (-b + sqrt(b^2 - 4*a*cc))/(2*a);
PmindBm = 10*log10(Pmin) + 30;
%% 5. Display results
[~,k] = min(abs(Pdbm+30));          % index of -30 dBm
fprintf('Responsivity R      = %.4f A/W\n', R);
fprintf('Thermal variance    = %.3e A^2\n', var_th);
fprintf('Dark variance       = %.3e A^2\n', var_dark);
fprintf('Shot variance(-30)  = %.3e A^2\n', var_shot(k));
fprintf('SNR at -30 dBm      = %.2f dB\n', SNR_dB(k));
fprintf('Pmin (SNR = 1)      = %.2f dBm (%.1f nW)\n', PmindBm, Pmin*1e9);
%% 6. Plots
subplot(2,1,1);
loglog(P, var_th*ones(size(P)), P, var_dark*ones(size(P)), P, var_shot, P, var_tot);
grid on; xlabel('P_{in} (W)'); ylabel('Noise variance (A^2)');
legend('Thermal','Dark','Shot','Total','Location','best');
subplot(2,1,2);
plot(Pdbm, SNR_dB); hold on; plot([-60 0],[0 0],'r--'); plot([PmindBm PmindBm],[-60
   70],'k:');
grid on; xlabel('P_{in} (dBm)'); ylabel('SNR (dB)');
```

**OUTPUTS:**

```
Command Window
Responsivity R       = 1.0625 A/W
Thermal variance     = 6.629e-13 A^2
Dark variance        = 1.602e-18 A^2
Shot variance(-30)   = 3.404e-16 A^2
SNR at -30 dBm       = 2.31 dB
Pmin (SNR = 1)       = -31.16 dBm (766.4 nW)
>>
```




**Reflection:**

1. The experiment helped me understand the impact of shot noise, dark current noise, and thermal noise on photodetector performance.
2. Thermal noise was found to be the dominant noise source at low optical power levels, significantly affecting receiver sensitivity.The MATLAB simulation demonstrated how SNR improves with increasing optical power and highlighted the importance of noise analysis in optical receiver design.

**CONCLUSION :**

The study shows how input optical power affects noise and SNR in a PIN photodetector receiver. Performance depends on the power level:

• Low power: thermal noise dominates, giving negative SNR and no reliable detection (below Pmin ≈ −31.16 dBm).

• Moderate power: the signal rises above the thermal noise and the SNR increases at 20 dB/decade. High power: SNR becomes large and shot noise grows linearly, approaching the thermal level near +2.9 dBm.

These results help in estimating the receiver sensitivity and in designing low-noise optical receivers.

**LABORATORY PERFORMANCE ASSESSMENT RUBRIC**

| **Objective** | **Assessment Criterion** | **Excellent – 4** | **Good – 3** | **Satisfactory – 2** | **Needs Improvement – 1** | **Weightage** |
|---|---|---|---|---|---|---|
| Develop the MATLAB program with correct constants and system parameters. | Program Setup and Parameter Definition | Defines all constants and parameters correctly with proper units and a clean script structure, independently. | Defines parameters correctly with minor unit or syntax errors and limited assistance. | Defines most parameters but needs considerable assistance or has unit errors (e.g., dBm/W, linear Fn). | Unable to set up the script or define the parameters. | 25 |
| Compute R, signal current, noise variances and SNR across the power sweep. | Simulation Execution and Data Acquisition | Computes R, Ip, all noise variances and SNR correctly over −60 to 0 dBm and records the values. | Computes all quantities with minor errors or omissions. | Computes some quantities with noticeable errors or inconsistencies. | Unable to run the simulation or record the values. | 25 |
| Obtain the noise and SNR plots and determine Pmin. | Graph Plo ing and Pmin Determination | Plots accurate log–log and semi-log curves with labels and reference lines; Pmin within 5% of the reference. | Plots the curves with minor errors; Pmin identified with small deviation. | Plots partial or poorly scaled curves; Pmin approximate. | Unable to plot the curves or determine Pmin. | 25 |
| Interpret noise regimes, SNR slopes and receiver sensitivity. | Analysis and Engineering Interpretation | Correctly identifies thermal- and shot-limited regimes, explains the SNR slopes and links results to receiver design. | Identifies the regimes with minor errors and reasonable interpretation. | Performs basic interpretation with noticeable errors or limited justification. | Unable to interpret the results. | 25 |